CS364 Semester Project

Project Proposal

Abstract

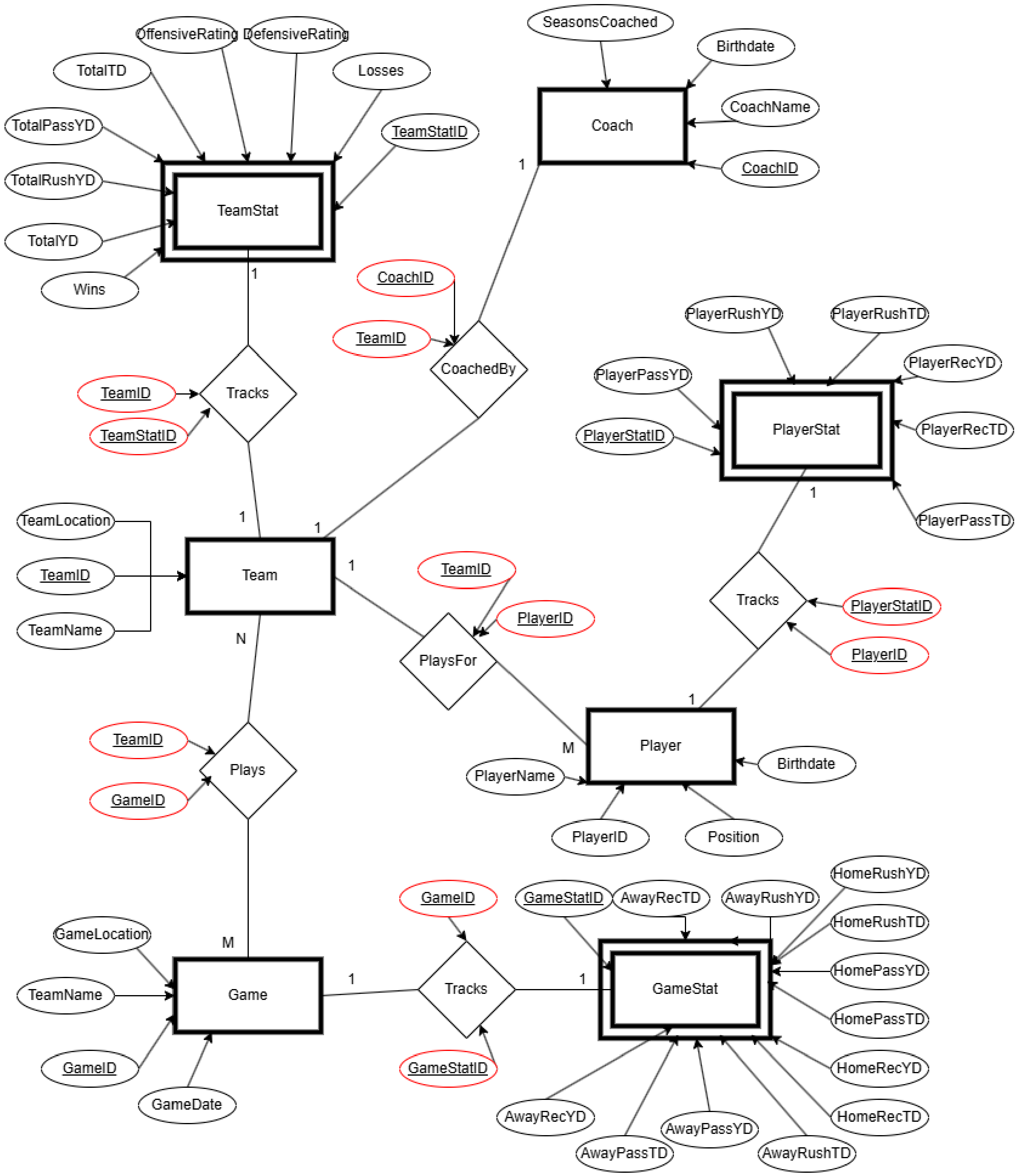
For this project, I want to create a Football Performance Database. This database will be designed to store and analyze detailed sports data from the NFL. The goal is to provide insights into team and player performance. This can help fans, analysts, and coaches evaluate which positions are excelling and which teams have the most efficient offense and defense. The database will store key statistics for players and teams, including performance metrics such as yards, touchdowns, and player efficiency ratings. It will also be capable of calculating advanced statistics like completion percentages, rushing efficiency, and turnover ratios that offer deeper football insights.

The system will allow users to run complex queries to assess the best-performing positions, offensive and defensive units, and much more. Users will also be able to compare performance across different time frames, such as single games or full seasons, to identify trends and patterns. The database will feature statistical breakdowns by player and team, enabling a more comprehensive analysis of game outcomes. This will hopefully also be insightful when trying to predict outcomes of future games based off past trends and statistics.

The Football Performance Database aims to be a valuable tool for football analysis. Users will be able to generate reports and visualizations that highlight performance trends over the course of a season or within individual games. This will hopefully allow its audience to examine football and sports from a more analytical, data-driven perspective, which could influence the evolution of strategies and decision-making within the game itself.

ER Diagram

We can create an ER diagram to help visualize what the Football Performance Database will look like from a conceptual schema standpoint. To represent the database, we will use 4 entities coach, team, player and game and 3 weak entities to track stats for team, player and game. Our main entities hold generic information about the entity while the statistic entities hold various football stats which we have interest in and would like to run queries on. We have various relationships as represented by diamonds in our diagram. The team entity has a relationship with virtually every part of our database and as you can see all the main entities have a relationship with their stat entities. We have IDs for all entities which will allow us to ensure easy access to specific players, coaches, teams or games. Foreign keys for relationships are represented in the ER diagram by red ovals and they are linked to the relationship.



Functionality

1. Player and Team performing tracking
    a. Store and update statistics for players and teams after each game
2. Game data tracking
    a. Store and update game specific statistics including scores, dates and teams
3. Player and team comparisons

    a. Allows user to compare individual's player performance and team performance
4. Report Generation
    a. Create reports that summarize player and team performance like trends over recent games

Some queries might include

- Top 5 players by position
- Best offensive teams
- Most yards by a player in a single game
- Player with the most touchdowns who is younger than the league average
- Team with the best record when playing at home

The possibilities for queries are endless as the user can ask any questions that align with the statistics being tracked in the database.

Stakeholders

There are many different types of users that may use the football performance database. This includes coaches, other analysts and fans. Coaches can use the database to analyze both team and individual performance. They can run queries to assess which positions or players are excelling and which areas need improvement. Coaches may focus on offensive and defensive rating, player efficiency stats and game performance to make strategic decisions. Analysts can use the advanced statistics stored in the database to try and evaluate trends and predict future performance. Like coaches they can use queries to compare players and teams based on historical data. Fans can use the database to track the performance of their favorite players and teams. Fans will primarily use the report and visualization to get quick insights into game stats. Another group that could use the database includes journalists who are reporting and writing about games. They can use the database to compare players and strengthen their takes by using the statistics found in the database.

Technological Requirements

This application will be implemented as a web-based platform to make it accessible to a wider audience, including fans, analysts, coaches and other stakeholders. For programming languages on the backend, we will be using Java for logic and data processing. For the front-end user interface, we will be using Python to create our website and gui. As a database for our platform, we will be using MySQL as the database management system, and we will be connecting to it via JDBC. This will allow us to run SQL queries from the backend to retrieve data. For sharing code, we plan to use email directly with all our code in a singular zip file. For this project I will be working alone because this is something I've wanted to do for a while, and I want to really understand it. My experience includes a good foundation for java and SQL. This will be my first time building a website using python which I have beginner level experience in. I plan to use Django to build my website using python.